

Single-Participant Assessment of Treatment Mediators

Strategy Description and Examples From a Behavioral Activation Intervention for Depressed Adolescents

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Determining the means by which effective psychotherapy works is critical. A generally recommended strategy for identifying the potential causal variables is to conduct group-level statistical tests of treatment mediators. Herein the case is made for also assessing mediators of treatment outcome at the level of the individual participant. Single-participant assessment of mediators requires documenting, for each participant, that treatment was received, that change occurred on the mediator and relevant clinical outcome measures, and that the change on the mediator happened at an expected time in the treatment protocol and prior to substantive change on the dependent variable. Data from four depressed adolescents who demonstrated remission following a behavioral activation intervention illustrate the use of the approach in assessing whether changes in activation level or negative thinking mediated the changes in depression. For two participants, increased activation appeared to be a mediator, whereas decreased dysfunctional thinking never emerged as a plausible mediator. It is concluded that single-participant assessment of mediators of treatment outcome offers a useful additional tool for determining possible mechanisms of action in effective psychotherapy.

Keywords: *mediators; behavioral activation; adolescent depression*

A recommended strategy for determining potential mechanisms of action in psychotherapy involves using statistical tests to identify mediators of treatment outcome (Holmbeck, 1997; Kazdin, 1999; Kraemer, Wilson, Fairburn, & Agras, 2002; Weersing & Weisz, 2002).¹ The logic and basic analytic strategy was described in a seminal paper by Baron and Kenny

(1986) and involves a set of conditions that need to be met for a variable to be considered a mediator. First, significant relationships must exist between the initial variable and the dependent variable, the initial variable and the hypothesized mediator, and the mediator and the dependent variable. Next, the mediator and initial variable are considered simultaneously (e.g., in a multiple regression model). Maintenance of the mediator-dependent variable relationship and elimination (ideally) or reduction of the initial variable-dependent variable relationship suggests the mediator accounts (at least in part) for the relationship between the initial variable and the dependent variable. To illustrate, consider recent data consistent with the hypothesis that decreased negative thinking may be the mechanism producing change in cognitive-behavior therapy (CBT) for youth depression (Kaufman, Rohde, Seeley, Clarke, & Stice, 2005). Ninety-three depressed, conduct disordered teens were randomized to CBT or a control condition. Both treatment group and change on the Automatic Thoughts Questionnaire (ATQ) significantly predicted change on the Beck Depression Inventory-II (BDI-II) and were significantly related to each other. However, when treatment group and ATQ change were considered simultaneously, the relationship between treatment group and BDI-II change was no longer significant, whereas the relationship between ATQ change and BDI-II change was maintained.

Although some additional/alternative methodological and statistical procedures have been suggested, this general approach has been widely endorsed for use in evaluating randomized clinical trial data (Kazdin, 1999; Kraemer et al., 2002; MacKinnon, Lockwood, Hoffman, West, & Sheets, 2002). However, like any analytic strategy, the determination of statistical mediators has strengths and limitations. One limitation is that the approach requires fairly large sample sizes. For example, Baron and Kenny (1986) recommend, as illustrated by Kaufman et al. (2005), calculating a series of regression models to test for significant bivariate relations followed by a multiple regression with the mediator and initial variable entered together. A standard rule of thumb for estimating number of cases in a multiple regression testing two predictors recommends 106 or more participants (Tabachnick & Fidell, 2001). If small relationships are expected or multiple mediators are to be tested, requiring that alpha-levels be adjusted, even larger sample sizes would be needed.

A related limitation is that the mediator is evaluated at the group level, which assumes a substantial amount of homogeneity of response to treatment across the sample. However, therapy likely operates through multiple mechanisms, such that any proposed mediator may apply to some but not

all clients. For instance, imagine that data from the treatment cell of a clinical trial consisted of a combination of subgroups: nonresponders, responders whose change is captured nicely by the mediated pathway, and responders for whom change does not correspond to the proposed mediated pathway. In such a scenario, having only a modest sample size would result in too little statistical power to identify a mediator, missing the subgroup whose data are captured by the mediated pathway and leading to the faulty conclusion that the treatment does not work via the proposed mechanism of action. This is an important consideration when the generally accepted idea that psychotherapy operates through multiple mechanisms is taken seriously. For instance, response to CBT for depression appears to involve both nonspecific and treatment-specific factors (Ilardi & Craighead, 1994). If the effects of psychotherapy are multiply determined, both within and across participants, analytic strategies are needed that are sensitive to and can reveal these processes.

Even when there is enough statistical power to identify a group-level mediator, a portion of the sample is still likely mischaracterized (i.e., any subgroup whose data may not be captured by the mediated pathway) or incompletely characterized. For instance, the relationships identified by Kaufman and colleagues (2005) were generally small-to-moderate in size, capturing a statistically significant amount of variance but leaving a majority of the variance unaccounted for. An important process was identified, but this process was not the only one relevant for understanding clinical improvement.

Based on the above considerations, it would appear useful if there were a strategy for assessing mediators that did not rely on a large sample size and was not tied to group-level statistical analyses. Such an approach would not replace statistical determination of group-level mediators but would provide an additional strategy for determining possible mechanisms of action in psychotherapy. In this article it is suggested that rigorously conducted, carefully planned and analyzed, single-participant designs can provide such an approach.

The term *single-participant* refers not to the overall number of participants but rather to the level of analysis, that is, that the results are evaluated at the level of the individual rather than the group. A single-participant design typically involves the participant being exposed to all, or multiple, levels of the independent variable, such that he/she serves as his/her own control, and the taking of repeated measures throughout the course of baseline and intervention phases to determine treatment effects (see Gaynor, Baird, & Nelson-Gray, 1999; Hayes, Barlow, & Nelson-Gray, 1999). Although single-participant designs have a long history in clinical research,

there is currently no detailed rationale or explicitly articulated strategy for using single-participant designs to study potential mechanisms of action in psychotherapy. Building on basic single-participant methodology and incorporating components standard to clinical outcome research generally, a logic and initial strategy for assessing mediators of treatment outcome at the single-participant level is presented. In brief, single-participant assessment of treatment mediators requires documenting that the participant: (a) received treatment; (b) improved during treatment; (c) showed change on the proposed mechanism of action, which occurred at a reasonably expected time given the treatment protocol; and (d) preceded a substantial portion of the clinical improvement. These steps are elaborated below.

The single-participant analysis of mediators begins with an assessment of the relationship between receiving treatment and outcome. A mechanism of action accounting for why therapy worked could not be reasonably identified in a client who did not get treatment. This obvious necessity requires documenting that the participant not only attended treatment sessions, but was also an active recipient of the treatment offered. Moreover, it must be determined that the treatment was provided to the participant as specified and with adequate skill. If either client compliance or therapist integrity is absent, treatment was not genuinely implemented, calling into question the appropriateness of trying to evaluate mechanisms of action to account for why the (unimplemented) treatment works.

Prior to trying to assess why treatment worked, it must also be convincingly demonstrated that positive change occurred during the time frame in which treatment was received. First, multi-method, multi-informant data from psychometrically sound instruments should document change from pretreatment to posttreatment. Next, the time course of change should be examined to determine when improvement occurred. For instance, in CBT for depression, a great deal of change occurs very early in treatment (Ilardi & Craighead, 1994), sometimes beginning before treatment starts (Gaynor et al., 2003). Such change could be misattributed if, for example, depression was measured only at intake and posttreatment and the mediator measured at the mid-point (Kazdin & Nock, 2003; Kraemer et al., 2002). Because of their use of repeated measurement at baseline and regular intervals throughout treatment, single-participant designs are especially suited to documenting the time course of change.

Having established that the client received the treatment and showed a positive outcome, the next step entails demonstrating that there was a change in the putative mechanism of action (i.e., the mediator). If there

were no change in the proposed mechanism of action, this could not reasonably be the active agent of change. For example, if during CBT for depression, symptoms improved, but there was no change in negative thinking, change in cognition could not be the mechanism of action for that individual. Moreover, the change in the mediator should occur at a generally expected point in the course of treatment—during or after events in therapy that target the proposed mediator for change.

The final step involves documenting that the change on the proposed mediator preceded a substantial amount of clinical improvement (Kazdin & Nock, 2003; Kraemer et al., 2002). If a variable is a mechanism of action accounting, at least in part, for why a therapy works, then change on this variable should precipitate symptom change. Demonstrating this temporal relationship is “absolutely vital in order to infer the operation of a mechanism of change” (Kazdin & Nock, 2003, p. 1121). Furthermore, if there is a change in the proposed mediator but no change on the outcome variable, the theorized mechanism is called into question. In addition, if the clinical improvement precedes the change on the proposed mediator, the latter change could not account for the clinical improvement. Relatedly, if change in the mediator merely coincides with clinical improvement, it may be epiphenomenal and not a mechanism. One identified limitation of the Kaufman et al. (2005) data was that the ATQ and BDI-II were taken simultaneously and only at intake and posttreatment, such that temporal precedence could not be determined. The importance of the time course of change on the mediator and dependent variable in drawing conclusions emphasizes the need to take repeated measures throughout the course of treatment to document the sequence in which symptom change occurs and to link this change to events occurring in treatment.

The present investigation illustrates single-participant assessment of mediators of treatment outcome with four depressed adolescents treated with a behavioral activation (BA) intervention. BA treatments for depression have received a great deal of attention following studies demonstrating their efficacy (see Dimidjian et al., 2006; Jacobson et al., 1996). The first large-scale study was Jacobson et al.’s (1996) dismantling trial showing BA to be as efficacious as a full CBT package. The data, however, failed to support activation as the mechanism of action in BA. Activation increased and dysfunctional thinking decreased across groups, indicating that the hypothesized mechanisms had been engaged. However, contrary to expectations, in BA early change in dysfunctional thinking predicted later change in depression, whereas in CBT early change in pleasant events predicted later change in depression. These findings are far from definitive, however,

because the assessment intervals were so broad (i.e., pre- to mid-treatment and mid- to posttreatment) as to be insensitive to temporal sequences of change occurring across smaller time frames (e.g., from week-to-week) that likely need to be explored to identify mediators (Kazdin & Nock, 2003; Kraemer et al., 2002).

Some have speculated that BA may represent a particularly efficient means of changing cognitions (Hollon, 2001). Whisman (1999) suggests that "an adequate test of the cognitive theory of change would involve repeated assessments of cognitive change and symptom change, to determine if cognitive change precedes symptom change in therapy" (p. 302). The BA parallel requires taking repeated measures to determine if a change in activation precedes symptom change. In the present study, these two competing mediators (activation and dysfunctional thinking) are evaluated at the level of the single-participant during BA treatment for depression. Conducting comparisons of competing mediators has been recommended but rarely occurs (Kazdin & Nock, 2003).

Mediators and moderators of treatment outcome are often discussed together. For this reason, the latter warrants some mention even though it is not the primary focus of this article. Moderators are pretreatment characteristics that influence treatment outcome (Baron & Kenny, 1986; Holmbeck, 1997; Kraemer et al., 2002). Thus, moderators identify subgroups for whom a treatment may be more or less effective. In the present study, potential moderators included pretreatment depression severity and duration, comorbidity, gender, age, and family structure (e.g., single- vs. two-parent family).

Method

Participants

Ten adolescents, recruited from the community via advertisements and fliers distributed to schools and health care professionals, were screened as part of the investigation. Four were not eligible for participation because their depressive symptoms were too mild. Six participants met inclusion criteria and were enrolled in the protocol. Two of these individuals dropped out following the first treatment session, the remaining 4 participants completed treatment. These 4 participants displayed at least moderate symptoms of depression, as assessed by the Diagnostic Interview for Children and the BDI-II. Exclusionary criteria included primary symptoms of obsessive compulsive disorder, psychotic disorders, autism spectrum disorder, or report of acute suicidality. No participants received additional psychotherapy

or pharmacotherapy. The therapist was a female doctoral student in clinical psychology who had completed graduate coursework and workshops in CBT and had 2 years of clinical experience using CBT techniques.

Experimental Design

A single-case A/B/B+C design was used for each participant, where A was baseline; B was psychoeducation, preparation, and generic behavioral activation; and C was values-based behavioral activation. Repeated measures were collected during baseline, prior to each treatment session, at 1-week posttreatment and follow-up. Several lengthier measures were administered only at pretreatment, posttreatment, and follow-up.

Procedure

Baseline Assessment: Phase A

Each participant attended at least three pretreatment meetings during which informed consent was obtained, a thorough assessment conducted, and baseline data collected. (Some participants completed repeated measures between assessment sessions resulting in more than three baseline data points.)

Treatment: Phases B and B+C

The treatment was values-based behavioral activation (VBBA). VBBA incorporated elements of behaviorally based treatments developed by Hayes, Strosahl, and Wilson (1999); Martell, Addis, and Jacobson (2001); and Lejuez, Hopko, and Hopko (2001). The approach was adapted for use with adolescents and codified into a delivery protocol (manual available from the authors). The therapy consisted of 12 one-hour meetings, with twice weekly meetings for the first 4 weeks followed by weekly meetings. As in other clinical trials for depressed youth, flexible use of one or two parent-child sessions were allowed, as needed, to facilitate implementation of the intervention.

Pre-VBBA (psychoeducation, preparation, and generic BA): Phase B. VBBA did not begin until after session 6. Sessions 1 through 6 provided an important lead into VBBA, but the focus and assigned homework involved activation that, for the most part, laid the groundwork for VBBA. For

instance, participants were assigned to read a psychoeducation pamphlet about BA, self-monitor daily activities and mood, fill out a values assessment form, complete an activities checklist, and develop a list of values-based activities. Generic activation prescriptions (e.g., related to exercise, sleep hygiene, and practice of mindful relaxation breathing) were given during this phase but were not explicitly linked to values.

The first session involved outlining the therapist's stance, providing psychoeducation, and building the therapeutic alliance. The therapist described her position as a consultant/coach whose job was to help the adolescent choose and develop a behavioral plan to help him/her pursue life goals and bring a sense of pleasure and pride. The objectives of the second session were to begin an idiographic functional analysis of the adolescent's depression. The adolescent was asked about current activities, attempts to alleviate depression, and, for homework, to self-monitor activities and mood. The third session involved psychoeducation about depression from a BA standpoint, stressing the interconnection of thoughts, feelings, and behaviors, and that if the participant changed his or her behavior, it would likely affect the others. Values were introduced in the fourth session as a way to enable the adolescent to focus his/her behavior on things that were most important to him/her. Part of the introduction involved presenting a values assessment form (adapted from Hayes et al., 1999) used to help the adolescent articulate and understand his or her values in the areas of family/caregiver relations, social relations, dating relations, learning/education, health/fitness/self-care, recreation/hobbies, citizenship, and spirituality. The fifth session began with a discussion of the values assessment form and the idea of translating values into activities. Once the adolescent had identified important values, the therapist and teen began formulating a list of values-consistent activities. These activities were then listed on a form, "Things I Can Do to Behave Consistently With My Values" (adapted from Lejuez et al., 2001). The adolescent also completed an activities checklist (modified from Clarke, Lewinsohn, & Hops, 1990), which was used in session 6 to help complete the "Things I Can Do" chart.

VBBA: Phase B+C. VBBA began after the sixth session during which the "Things I Can Do" chart was finalized. The adolescent was instructed to try and engage in the selected activities every day (if feasible) and to record this engagement using the "Things I Can Do" chart. The adolescent was encouraged to review this form each morning and evening as a prompt, but also to engage in values-based behaviors that were not on the list, as opportunities were spontaneously identified, and to then add them to the

list. In the remaining sessions, the therapist and youth reviewed activities completed since the last session, changing or adding activities as dictated by the teen's circumstances and experiences. If the teen encountered problems engaging in an activity, he or she and the therapist worked together to try to devise solutions or to find alternative, relevant activities.

Measures

Receipt of Treatment

Attendance was the number of sessions attended. *Therapist adherence and competence* and *client compliance* were derived from a treatment integrity measure completed at the end of each session by the therapist and, for 25% of sessions, an independent, trained, doctoral student coder. The coded videotapes were selected quasi-randomly to ensure an even distribution across participants (three tapes for each participant) and sessions (four from sessions 1 through 4; four from sessions 5 through 8; and four from sessions 9 through 12).

Adherence refers to whether the therapist implemented the treatment as specified in the protocol, whereas *competence* involves the skillfulness of the implementation (Waltz, Addis, Koerner, & Jacobson, 1993). Seven items tapped both adherence and competence. All of these items have parallels in or were taken directly from existing CBT/BA integrity measures (Gollan, Gortner, Koerner, & Jacobson, 1995) and were scored from 0 (*poor*) to 6 (*excellent*). Five of the items tapped general behavior therapy skills, such as collaboratively establishing and following an agenda, reviewing and assigning homework, demonstrating understanding of the client, reviewing depressive symptoms, and asking for feedback and ensuring client understanding. These items are referred to collectively as behavior therapy (BT) competence. Two items were specific to behavioral activation: emphasis placed on the function of behavior in context and an overall rating of the clinician's skill as a behavioral activation therapist. These items are referred to as behavioral activation (BA) competence.

Although the BT and BA competence subscales captured adherence to behavior therapy and behavioral activation principles in general, they did not address adherence to the VBBA protocol in particular. For that reason, one to five session-specific therapist items were included. These items assessed whether therapist behaviors that should have occurred in a particular session according to the VBBA protocol did indeed occur. These are referred

to as VBBA adherence. For instance, did the therapist present the BA rationale using the handout (session 2), explain the use and purpose of the Daily Activity Log (session 2), use the information collected to make the BA rationale idiographic (session 3), introduce and discuss values (session 4), explain translating values into action (session 5), develop an idiographic values-based activities list (session 6), review activities from the previous week and formulate an activity plan for the next week, examine likely outcomes of different activities, and encourage engagement in specific activities (sessions 7 through 12)? All were scored from 1 (*not at all*) to 6 (*extensively*).

Client compliance was derived from three items focused on the teen's engagement in extra session activities. The items involved ratings of the amount and thoroughness of homework completed and of the overall pursuit of treatment goals based on descriptions of activities occurring between sessions. These items were scored from 0 (*not at all*) to 4 (*completely*).

Clinical Improvement

The computerized NIMH Diagnostic Interview Schedule for Children (DISC; Shaffer, Fisher, Lucas, Dulcan, & Schwab-Stone, 2000) was used to assess psychopathology across the major diagnostic categories of the *DSM-IV*. The modules for major depression, dysthymia, bipolar disorder, generalized anxiety, social anxiety, conduct disorder, substance use disorder, and attention-deficit/hyperactivity disorder (ADHD) were administered at pretreatment, posttreatment, and follow-up. The widely used 21-item self-report BDI-II (Beck, Steer, & Brown, 1996) was administered at pretreatment, posttreatment, follow-up, and prior to each session. The BDI-II has been used on a session-by-session basis in youth depression trials (see Brent et al., 1997) and the following interpretive ranges recommended: 0 to 9 = nondepressed; 10 to 15 = mild depression; 16 to 23 = moderate depression; and > 24 = severe depression (Roberts, Lewinsohn, & Seeley, 1991). The Child Behavior Checklist-Parent Rating Form (CBCL; Achenbach, Howell, Quay, & Conners, 1991) is composed of 118 items assessing internalizing and externalizing behaviors of the child; *T* scores below 60 are considered in the normative range. A 15-item CBCL-Depression score was also calculated (as described by Clarke, Lewinsohn, Hops, & Seeley, 1992) to provide a measure of depression-specific symptom change from the perspective of a parent.

Potential Mediators

The ATQ consists of 30 items measuring dysfunctional thinking and is considered a useful tool in assessing change in cognitions related to psychotherapeutic intervention (Hollon & Kendall, 1980). The ATQ was administered at pretreatment, posttreatment, and follow-up. The seven items comprising the Negative Self-Concepts and Negative Expectations (NSNE) subscale of the ATQ were completed prior to each session. The COPE (Carver, Scheier, & Weintraub, 1989) is a 60-item scale assessing how individuals respond when confronted with difficult or stressful events. Two subscales, active coping and behavioral disengagement, were the focus of the present investigation. The eight items, scored on a 1- to 4-point Likert scale assessing behavioral disengagement (e.g., "I just gave up trying to reach my goals") and active coping (e.g., "I took action to try to get rid of problems"), were completed at pretreatment, posttreatment, follow-up, and prior to each session. The instructions and item wordings were modified slightly to represent current coping responses rather than general reactions to stressful events. COPE activation was determined by subtracting the behavioral disengagement total from the active coping total. As such, positive scores indicate greater activation than disengagement.

Self-monitored activity level was measured using the "Things I Can Do to Behave Consistently With My Values" form. This form was adapted from several forms used by Lejuez et al. (2001) and contained the list of values-based activities generated by the therapist and teen in rows, and the days of the week in columns. The participant was asked to circle a preprinted *Y* (for *yes*) or *N* (for *no*) each day to indicate whether he or she had engaged in that activity and to make a mood rating from 1 (*very irritable or sad*) to 10 (*very positive*). A therapist rating of client compliance was taken from the item on the treatment integrity measure assessing the therapist's impression of how active the client was in pursuit of treatment goals based on the in-session verbal review of activities since the previous session.

Results

Participant PO: Strong Evidence for Activation as Mediator

PO was a Euro-American female who resided with her mother, stepfather, and three younger siblings. Her mother had a college education and was employed part-time. Her stepfather had a high school education and was

self-employed. PO was assessed at 14 years, 4 months, and met intermediate criteria for a current major depressive episode. Symptom duration was 6 months.

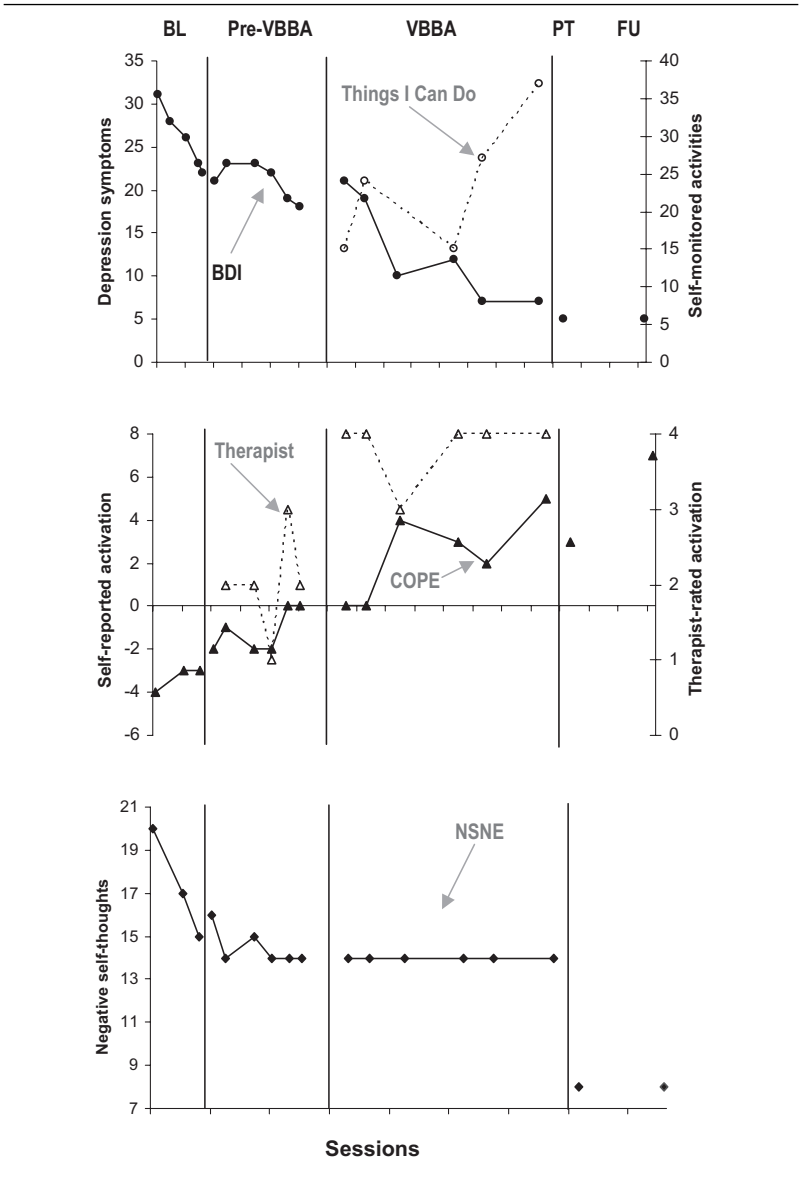
Receipt of treatment. PO attended 12 individual sessions during 12 weeks. After 4 sessions, an added mother-child meeting was held to discuss expected behaviors of PO in the home. Also, to facilitate activation, the therapist met individually with PO's mother to discuss normal expectations of adolescent behavior and appropriate freedoms, privileges, and parental monitoring. Follow-up assessment occurred 1 month posttreatment.

The correlation between coder and therapist ratings indicated significant agreement ($n = 37$, $r = 0.86$, $p < .000$). BT competence ($n = 15$) averaged 5.60 ($SD = 0.63$) for the coder and 5.73 ($SD = 0.46$) for the therapist; BA competence ($n = 6$) averaged 5.33 ($SD = 0.52$) for the coder and 5.17 ($SD = 0.75$) for the therapist; and VBBA adherence ($n = 7$) averaged 6.0 ($SD = 0.0$) for the coder and 5.86 ($SD = 0.38$) for the therapist. These data suggest very good to excellent treatment integrity, which was maintained when averaged across all of PO's sessions: BT competence ($n = 60$) averaged 5.63 ($SD = 0.88$); BA competence ($n = 24$) averaged 5.29 ($SD = 0.55$); and VBBA adherence ($n = 38$) averaged 5.61 ($SD = 0.75$). Client compliance ($n = 9$) averaged 3.11 ($SD = 1.17$) and 3.00 ($SD = 0.87$) according to the coder and therapist, respectively, suggesting she was generally compliant with treatment. The average across all sessions was similar ($n = 33$, $M = 3.06$, $SD = 1.14$, $MODE = 4.0$).

Clinical improvement. PO demonstrated a substantial change in depressive symptoms from pretreatment to posttreatment across measures and informants. PO no longer met DISC criteria for a major depressive episode at posttreatment and her BDI-II score fell in the nondepressed range. CBCL depressive symptoms decreased from 11 to 2. In addition, her pretreatment internalizing and externalizing scores were both in the clinical range ($T_s = 69$ and 66, respectively), whereas posttreatment scores were below (internalizing $T = 52$) or at normative cutoffs (externalizing $T = 60$).

As is apparent in the top panel of Figure 1, the repeated measures assessment of depression reveals evidence of a pretreatment change of about 10 BDI-II points. This declining trend on the BDI-II leveled off at the end of baseline and remained relatively high and stable (BDI-II ~ 20) across the pre-VBBA phase and initial portion of VBBA. As such, it was possible to observe significant clinical improvement during the course of treatment that did not seem to represent a simple continuation of the downward slope occurring during baseline.

Figure 1
Repeated Measures Data for Participant PO



Note: BDI-II = Beck Depression Inventory-II; NSNE = Negative Self-Concepts and Negative Expectations.

Change on potential mediators and the timing of mediator change. PO demonstrated a decrease in negative thinking from a pretreatment ATQ of 79 to a posttreatment of 37. Likewise, her COPE activation score increased from a -3 to a 3. More important, as is apparent in the top and middle panels of Figure 1, PO clearly increased her activation level during VBBA. PO self-monitored engaging in 15, 24, 15, 29, and 42 activities in the five intervals for which she returned her "Things I Can Do" forms (top panel, dashed line). This activation was also apparent in the therapist ratings (middle panel, dashed line), which increased and remained high only during VBBA. Indeed, 5/6 VBBA data points show no overlap with the prior phase. COPE activation, which showed modest improvement prior to VBBA, also rose substantially during VBBA, with the final four data points falling outside the range of the prior phase. With respect to the alternative mediator, no changes on the NSNE were observed during this time period (lower panel). Instead, NSNE change was apparent only at posttreatment and 1-month follow-up.

To further examine the pattern of change, we calculated ipsative z scores for both the COPE and the therapist-rated activation score for each treatment session across the pre-VBBA and VBBA phases. Ipsative z scores are calculated by subtracting each individual session score from the participant's average score on that measure and then dividing by the participant's standard deviation (see Mueser, Yarnold, & Foy, 1991). The sign of the resulting ipsative z score indicates whether the score for that session was higher, lower, or equal to the average for that participant and, thus, can provide some information about when in treatment change began to occur (i.e., based on changes in the sign of the ipsative z score). To examine the pattern of change, we coded ipsative z scores suggesting change in the therapeutic direction with a 1, and coded those that were not in the therapeutic direction with a 0. On the COPE, the results were: pre-VBBA = 000000 and VBBA = 001111, whereas the therapist-rated activation results were pre-VBBA = x00000 and VBBA = 110111. As with the visual analysis of Figure 1, these data show that increased activation was noted only during the VBBA phase. The timing of the increased activation is consistent with what would be expected if it were due to the introduction of VBBA. That is, that VBBA appeared to lead to the change in the proposed mediator.

The relationship between mediator change and clinical improvement. As is apparent in Figure 1 (top panel, solid line), the observed within-treatment improvement on the BDI-II occurred only during VBBA. Moreover, this BDI-II change followed a sustained period during which there was little to

no improvement. Thus, both an increase in activation and a decrease in depressive symptoms were associated with VBBA. Moreover, the data suggest a temporal link wherein activation preceded changes in depressive symptomatology. Examination of the top panel of Figure 1 (solid line) shows that it was only in session 9 that substantive change appeared on the BDI-II and that this BDI-II change was preceded by a significant amount of values driven activation according to client self-monitoring of activities (upper panel, dashed line) and therapist ratings of active pursuit of treatment goals (middle panel, dashed line). In addition, high levels of activation were maintained or achieved on all three activation measures prior to the continued BDI-II decline observed in sessions 11, 12, and posttreatment.

To further document the pattern of change, ipsative z scores (Mueser et al., 1991) were calculated for each BDI-II across the pre-VBBA and VBBA phases. As noted earlier, ipsative z scores suggesting change in the therapeutic direction were coded with a 1, those that were not in the therapeutic direction with a 0. These data (pre-VBBA = 000000 and VBBA = 001111) make clear that the within-treatment decrease in depression was noted only during the VBBA phase. Furthermore, the coded ipsative z scores also showed that therapist-rated activation (identified in the first two VBBA sessions) preceded the changes in depression.

The binary coding of ipsative z scores illustrates when change in a positive direction occurred but does not speak to the magnitude of the change. However, it is possible to calculate an intra-individual critical difference value that can be used to determine if there is a statistically significant difference between pairs of ipsative z scores (Mueser et al., 1991). For PO, we assessed whether the first positive ipsative z score on the therapist-rated activation measure and BDI-II were significantly different from the preceding score at the $p < .05$ level. The resulting critical values that needed to be exceeded were $(1.64[2(1 - .87)]^{1/2}) = 0.82$ for the BDI-II and $(1.64[2(1 - .47)]^{1/2}) = 1.69$ for therapist-rated activation (see Meuser et al., 1991, for a complete discussion of critical value derivation). In both cases, the critical difference criterion for statistical significance was exceeded (BDI-II = 1.47, therapist-rated activation = 1.83), indicating that introduction of VBBA was followed by a statistically significant increase in therapist-noted activation, which preceded a statistically significant decrease in depressive symptoms. Thus, both visual/graphical and statistical analyses support the idea that VBBA produced an increase in PO's activity level, which preceded and mediated the within-treatment symptomatic improvements.

Participant SR: Plausible Evidence for Activation as Mediator

SR was an 18-year-old Euro-American male residing with his parents, both of whom had a high school education. His father was retired and receiving disability, whereas his mother worked part-time. SR met intermediate criteria for a current major depressive episode on the DISC, with a symptom duration of 4 months. He also met for additional diagnoses of conduct disorder, ADHD (intermediate), and marijuana dependence. At the time of the assessment, SR had failed grade 12 due to excessive absences and was in the process of repeating this grade. In addition, he had been sentenced to 12 months of probation following an arrest for shoplifting and possession of marijuana.

Receipt of treatment. SR received 12 individual sessions and one parent-child session during 12 weeks. The family session occurred after the fourth individual session. At this meeting, SR and his parents developed a contract that specified behaviors expected of SR and resulting consequences for compliance and noncompliance. Follow-up was 1 month after treatment.

The correlation between coder and therapist ratings indicated significant agreement ($n = 41$, $r = 0.87$, $p < .000$). BT competence ($n = 15$) averaged 5.00 ($SD = 0.93$) for the coder and 5.80 ($SD = 0.56$) for the therapist; BA competence ($n = 6$) averaged 5.00 ($SD = 0.89$) for the coder and 5.50 ($SD = 0.55$) for the therapist; and VBBA adherence ($n = 11$) averaged 5.18 ($SD = 1.47$) for the coder and 4.91 ($SD = 1.76$) for the therapist. These data suggest very good treatment integrity, which was maintained when averaged across all of SR's sessions where BT competence ($n = 60$) averaged 5.55 ($SD = 0.70$), BA competence ($n = 24$) averaged 5.00 ($SD = 0.72$), and VBBA adherence ($n = 39$) averaged 4.74 ($SD = 1.83$). Client compliance ($n = 9$) averaged 1.89 ($SD = 1.83$) and 1.44 ($SD = 1.74$) according to the coder and therapist, respectively. Client compliance across all sessions similarly suggested modest compliance ($n = 33$, $M = 1.21$, $SD = 1.49$).

Clinical improvement. SR reported marked change in depressive symptoms, which was modestly corroborated. At posttreatment SR no longer met DISC criteria for major depression and his BDI-II score fell in the nondepressed range. A more modest change in depressive symptoms was noted on the CBCL, which decreased from 14 to 9. SR still met criteria for conduct disorder, ADHD (intermediate), and marijuana dependence (intermediate). Significantly lower remission rates for comorbid conduct disorder than depression have been reported in a clinical trial of CBT (Rohde,

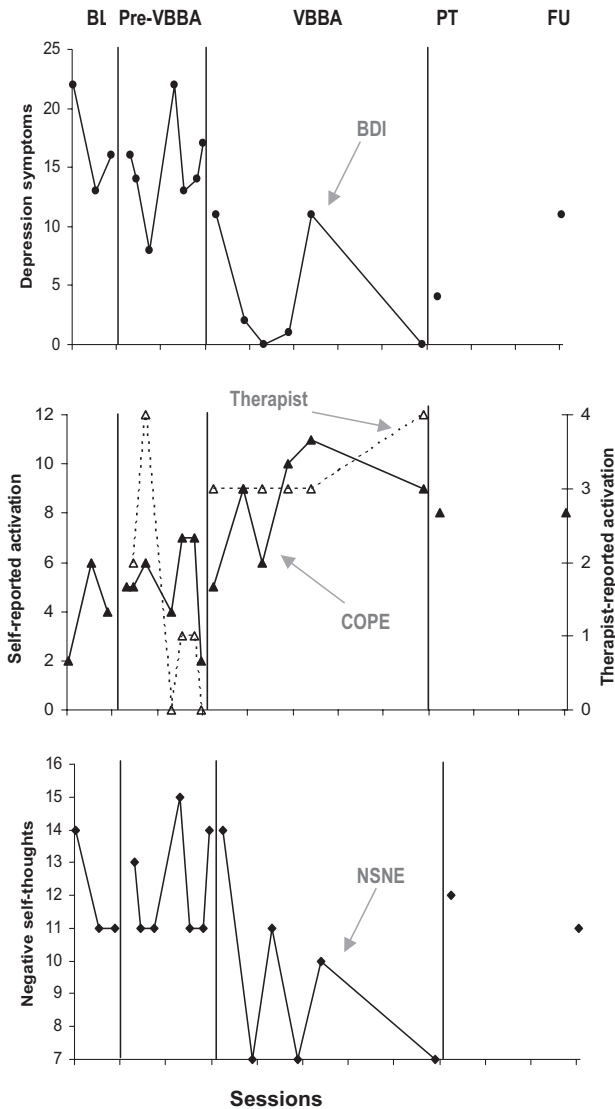
Clarke, Mace, Jorgensen, & Seeley, 2004). In addition, changes in these disruptive behavior and substance use diagnoses are more difficult to document in relatively brief treatment protocols because the *DSM* criteria for these disorders sample behavior during the past 6 to 12 months. However, no significant changes from pretreatment to posttreatment in internalizing symptoms ($T_s = 69$ to 66) or externalizing symptoms ($T_s = 63$ to 62) were noted on the CBCL, suggesting the lack of change is not simply an artifact.

Compared to his initial BDI-II, SR reported a decrease in depressive symptoms at the second baseline assessment point, which slightly increased at assessment point 3. Looking at SR's baseline data in the context of his pre-VBBA data does not suggest a downward trend, but instead weekly variation in functioning centered around the moderately depressed range (see Figure 2, top panel). Thus, according to SR's report, substantial clinical improvement occurred during the course of treatment.

Change on potential mediator and timing of mediator change. SR reported a change in COPE activation as his score doubled, going from a pretreatment score of 4 to a posttreatment score of 8. SR did not endorse a great deal of negative thinking at pretreatment (ATQ = 46) and reported a minor increase at posttreatment (ATQ = 59); however, both scores fell within the normative range.

SR showed no consistent changes in COPE activation or negative thinking during the pre-VBBA phase of treatment. He completed few formal homework assignments and demonstrated limited and declining pursuit of treatment goals during this time period. Consistent with his failure to complete most of the formal homework, SR did not return any "Things I Can Do" charts during VBBA. However, he was able to articulate actively pursuing treatment goals to a degree sufficient to be captured in therapist session ratings, which showed an immediate, marked increase during VBBA (see Figure 2, middle panel, dashed line). SR described engaging in activities related to completing his probation requirements (e.g., volunteering at Habitat for Humanity), regularizing his sleep, improving his appearance, spending time with his nieces and brother-in-law, exercising, reconnecting with former friends, and abstaining from alcohol and marijuana. SR's COPE score also increased during VBBA with 4/6 data points falling outside the range established in the pre-VBBA phase. The pattern of change on the COPE and therapist-rated activation according to the coded ipsative z scores further supports the suggestion that consistent increases in activation occurred only during VBBA. On the COPE the results were: pre-VBBA = 0000110 and VBBA = 010111, whereas the therapist-rated activation

Figure 2
Repeated Measures Data for Participant SR



Note: BDI-II = Beck Depression Inventory–II; NSNE = Negative Self-Concepts and Negative Expectations.

results were pre-VBBA = x010000 and VBBA = 111111. SR also demonstrated a substantial decrease in negative thinking that was exclusive to the VBBA phase (see Figure 2, bottom panel). The timing of the activity level and NSNE changes suggest they were plausibly the result of VBBA. Moreover, the fact that SR was often noncompliant with treatment and only demonstrated change following the increase in compliance during VBBA further suggests that the treatment (and not nonspecific factors, such as simply spending time with a supportive listener) led to these changes.

The relationship between mediator change and clinical improvement. As shown in Figure 2 (top panel), depressive symptoms declined to a low level only during VBBA, where 4/6 data points fell outside the range of the prior phases. Thus, VBBA resulted in increased activation (across measures), decreased negative thinking (on the NSNE repeated measure), and decreased depressive symptoms. But did activation or negative thinking mediate the relationship between VBBA and symptom change? In the second VBBA session, the BDI-II reached its most improved level up to that point in treatment and, for the first time, was clearly outside the range established by the preceding phases. In the previous session, the therapist noted an increase in activation (see Figure 2, middle panel, dashed line), which was sustained, providing evidence of a temporal linkage. There was no temporal precedence established for the COPE or for the NSNE, and the lack of consistency between the NSNE and posttreatment ATQ further calls this potential mediator into question. Given that the only evidence of temporal precedence was the therapist rating of activation, it is important to determine the reliability of this rating, which we were able to do because this session was one of the coded sessions for SR. Indeed, the coder rated SR's active pursuit of treatment goals in the preceding week even higher (i.e., 4) than the therapist (i.e., 3), further documenting that a substantial increase in activation preceded depression change.

The coded ipsative z scores further the suggestion that positive change on the BDI-II occurred primarily during VBBA (pre-VBBA = 0010000 and VBBA = 011101) and that therapist-rated activation (which was identified in the first VBBA data point) preceded the BDI-II change. Furthermore, the difference between the first therapist-rated activation score during VBBA and its predecessor (difference = 2.11) exceeded the critical difference for statistical significance, $(1.64[2(1 - .124)]^{1/2}) = 2.03$. The first positive BDI-II compared to its predecessor (difference = 1.72) was also significantly different at the $p = .05$ level, critical difference, $(1.64[2(1 - 1.05)]^{1/2}) = 1.72$. Thus, both visual/graphical and statistical analyses support the conclusion

that VBBA produced a marked increase in activity level, which predated the within-treatment symptomatic improvements.

Overall, the temporal sequence was not as clear, and the change in depressive symptoms not as consistent for SR as for PO. However, the available data suggest that the clinical improvement appeared attributable to VBBA, and that the most plausible mechanism was increased activation. Follow-up data also indirectly support this conclusion. At follow-up, the BDI-II had increased to near baseline levels. During this time, SR had several significant setbacks in areas that were of focus in VBBA—he was once again estranged from his brother-in-law and had pending charges for cocaine possession.

Participant ZA: Treatment Increased Activation, but Activation Did Not Mediate the Observed Symptom Change

ZA was an African-American female who resided with her mother and eight siblings and step-siblings. Her mother had a high school education and was employed part-time. She was assessed at the age of 12 years, 7 months, with the DISC and met criteria for a current major depressive episode with a symptom duration of 7 months. At the time of the assessment she was attending summer school as a result of failing seventh grade.

Receipt of treatment. ZA attended 12 sessions during 9 weeks. Follow-up assessment occurred 3 months after treatment. The correlation between coder and therapist ratings ($n = 49$, $r = 0.90$, $p < .000$) indicated significant coder-therapist agreement. BT competence ($n = 15$) averaged 5.47 ($SD = 0.52$) for the coder and 5.27 ($SD = 0.80$) for the therapist; BA competence ($n = 6$) averaged 4.33 ($SD = 1.37$) for the coder and 4.33 ($SD = 0.52$) for the therapist; and VBBA adherence ($n = 8$) averaged 5.00 ($SD = 1.77$) for the coder and 4.87 ($SD = 1.73$) for the therapist. These data suggest the treatment was delivered with good integrity, which was maintained when averaged across all of ZA's sessions where BT competence ($n = 60$) averaged 5.45 ($SD = 0.65$), BA competence ($n = 24$) averaged 4.50 ($SD = 0.51$), and VBBA adherence ($n = 39$) averaged 4.74 ($SD = 1.67$). Client compliance ($n = 6$) averaged 2.00 ($SD = 1.55$) and 2.67 ($SD = 2.07$) according to the coder and therapist, respectively. Across treatment ($n = 33$) average compliance was 2.48 ($SD = 1.27$), indicating ZA was some (2.0) to mostly (3.0) compliant with treatment.

Clinical improvement. ZA demonstrated a substantial change in depressive symptoms. At posttreatment she no longer met DISC criteria for a

major depressive episode and her BDI-II score was in the nondepressed range. On the CBCL, her pretreatment internalizing ($T = 73$) and externalizing ($T = 66$) scores decreased to 33 and 40, respectively. CBCL depressive symptoms fell from 17 to 1.

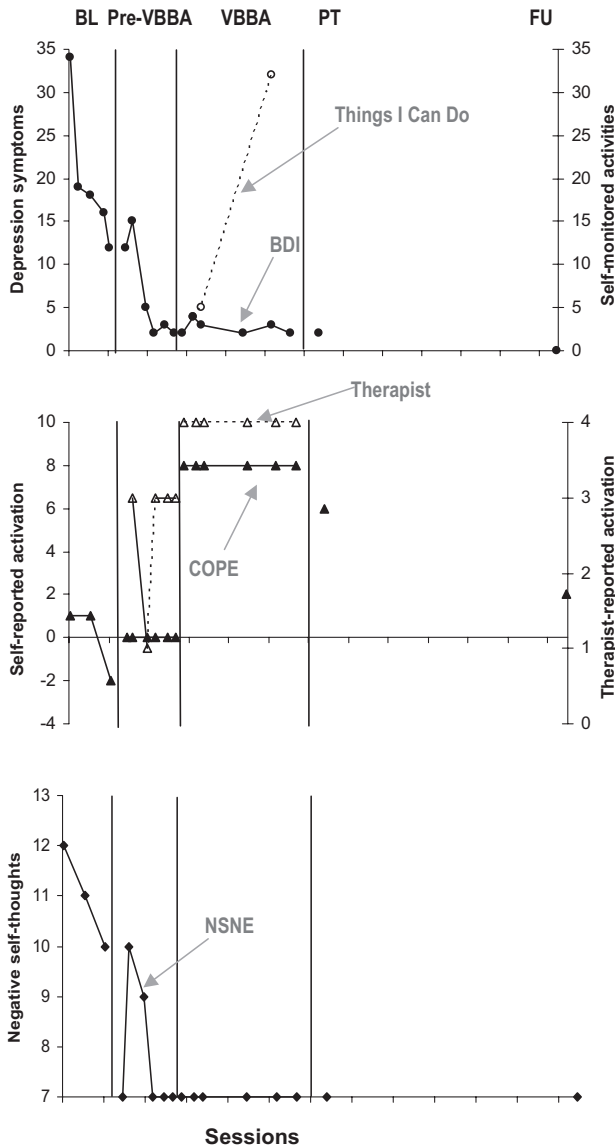
As Figure 3 (top panel, solid line) makes apparent, ZA reported a marked decline (20 BDI-II points) in depressive symptoms during baseline. Her depressive symptoms stabilized at a modestly elevated level across the final baseline point and first few weeks of treatment, such that the remaining clinical improvement occurred within the course of treatment.

Change on potential mediators and timing of mediator change. Despite not showing a great deal of negative thinking at pretreatment ($ATQ = 59$), ZA reported a decrease at posttreatment ($ATQ = 39$). On the NSNE subscale, ZA demonstrated a declining baseline, which bottomed out prior to the first treatment session, then rose in the second treatment session and returned to its lowest level by session 4 (see Figure 3, bottom panel). Thus, the change on the NSNE clearly preceded VBBA.

Improvement was noted in COPE activation, which increased from a -2 at pretreatment to a 6 at posttreatment. In addition, as is apparent in Figure 3 (middle panel, dashed line), whereas ZA was generally compliant with early treatment goals during pre-VBBA, she did not register this as a change in activation on the COPE (middle panel, solid line). However, introduction of VBBA clearly led to an increase in activation for ZA that was apparent on all measures. ZA completed and returned three "Things I Can Do" forms documenting completion of 4, 5, and 32 activities, respectively, in the domains of recreation, family, and social life. There were also immediate and consistent level changes on the COPE and in therapist-rated activation (see Figure 3), such that all VBBA ipsative z scores on these measures were coded 1, compared to none of the pre-VBBA z scores. In addition, the NSNE had reached its floor three sessions before the changes in activation; thus, whereas the role of changes in thinking in contributing to (or setting the stage for) increased activation cannot be ruled out, the most proximal change prior to the increased activation was the introduction of VBBA. That is, the greater activation appeared most likely due to VBBA.

The relationship between mediator change and clinical improvement. Examination of ZA's session-by-session BDI-II data (see Figure 3, top panel, solid line) suggests that the significant within-treatment symptom reduction occurred by the third session, well before introduction of VBBA. The therapist ratings of activation (middle panel, dashed line) indicated that she was mostly compliant with early treatment goals, which coincided with

Figure 3
Repeated Measures Data for Participant ZA



Note: BDI-II = Beck Depression Inventory-II; NSNE = Negative Self-Concepts and Negative Expectations.

the decrease in symptoms and negative self-thoughts. However, no clear temporal sequence was apparent and ZA did not report a change in COPE activation during this time. Thus, whereas pre-VBBA may have contributed to the change in symptoms, it is unclear as to whether this was the case and, if so, how it contributed. As such, the data appear most consistent with those from a subgroup of depressed teens who demonstrate large symptom changes early in treatment across intervention modalities (Gaynor et al., 2003).

Participant NM: Failure to Reliably Alter Activation Mediator

NM was a mixed ethnicity (Korean-American and Euro-American) male who initially resided with his father, step-mother, and four step-siblings. By the end of treatment, NM resided solely with his father. His father had a high school education and initially had seasonal employment, but by the end of treatment was unemployed. NM was assessed at 15 years, 2 months, and diagnosed with a current major depressive episode (symptom duration 14 months) and ADHD.

Receipt of treatment. NM began treatment after a 4-week baseline because an extra assessment session was required. He received 12 sessions during 16 weeks and two father-child sessions (between individual sessions 2 and 3, and 4 and 5). The goal of these meetings was to discuss NM's desire to spend more time with his father, to encourage NM's father to devote more time to his son, and to discuss possible barriers. NM completed follow-up 2 months after treatment completion.

The coder and therapist ratings were moderately to strongly correlated ($n = 36$, $r = .68$, $p < .001$). BT competence ($n = 15$) averaged 4.87 ($SD = 1.25$) for the coder and 5.73 ($SD = 0.46$) for the therapist; BA competence ($n = 6$) averaged 4.33 ($SD = 0.52$) for the coder and 5.17 ($SD = 0.75$) for the therapist; and VBBA adherence ($n = 7$) averaged 4.29 ($SD = 1.80$) for the coder and 4.43 ($SD = 2.37$) for the therapist. Despite the higher therapist competence ratings, even the coder ratings suggested quite acceptable competence and considerable adherence. The therapist ratings were consistent when averaged across all of NM's sessions where BT competence ($n = 60$) averaged 5.72 ($SD = 0.49$), BA competence ($n = 24$) averaged 5.17 ($SD = 0.48$), and VBBA adherence ($n = 37$) averaged 4.16 ($SD = 1.67$). Client compliance ($n = 9$) averaged 1.50 ($SD = 1.41$) and 1.44 ($SD = 1.24$) according to the coder and therapist, respectively, and across treatment ($n = 33$) averaged 1.06 ($SD = 0.93$), suggesting little treatment compliance.

Clinical improvement. NM reported marked change in depressive symptoms, which were somewhat corroborated. At posttreatment he failed to meet criteria for a major depressive episode (and ADHD), and his BDI-II fell in the nondepressed range. Moreover, CBCL depressive symptoms decreased from 13 to 7. However, overall internalizing symptoms remained elevated at posttreatment ($T = 69$) whereas externalizing symptoms fell just below the clinical cutoff ($T = 59$).

NM reported considerable pretreatment change as his initial BDI-II fell in the severe range, but dropped to below the clinical cutoff by the last baseline assessment point. The first treatment session saw a rise to the moderate range, which dropped again by the start of session 2 and remained at a low level thereafter.

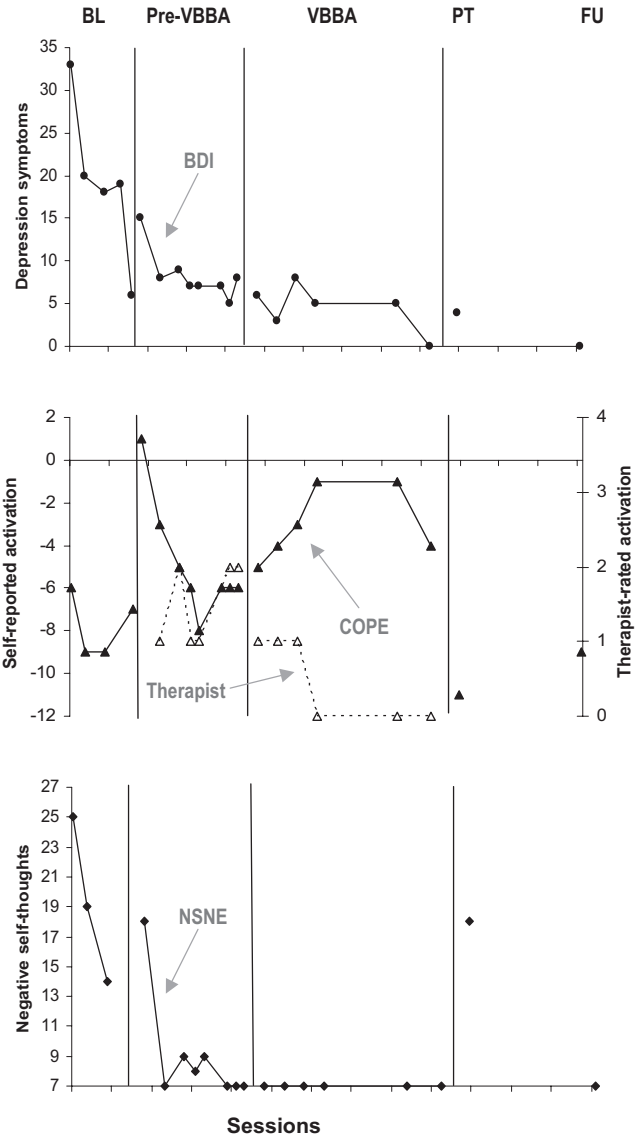
Change on potential mediators and timing of mediator change. NM reported elevated negative thinking at pretreatment ($ATQ = 76$), which modestly improved at posttreatment ($ATQ = 60$). However, his COPE activation score worsened from pretreatment to posttreatment, going from a -7 to a -11 . Figure 4 reveals a decreasing baseline on the NSNE (bottom panel) and stable low activation on the COPE (middle panel, solid line). The COPE reached its highest point at the first therapy session, which was followed by a drop to baseline levels that was maintained during pre-VBBA. The declining NSNE baseline trend was reversed at treatment session 1; however, a substantial decline occurred by session 2, which was maintained thereafter, well before VBBA began. During VBBA, NM reported a modest increase in activation relative to baseline and the majority of the previous phase; however, disengagement was always greater than active coping, and the therapist ratings contradict NM's self-report, suggesting a decrease in activation during VBBA. NM failed to return any "Things I Can Do" forms, providing no self-monitoring data. Thus, VBBA did not appear to lead to reliably increased activation.

The relationship between mediator change and clinical improvement. The vast majority of within-treatment depression change occurred in the interval from session 1 to session 2 and seemed to represent a continuation of a change process that began during baseline. There was no evidence that activation, and particularly VBBA, or changes in negative thinking accounted for the clinical improvement. Like ZA, NM's data appear most consistent with a subgroup of rapid early responders.

Moderators

Because of the small sample size and the fact that depression declined substantially for all participants, it is not possible to identify pretreatment

Figure 4
Repeated Measures Data for Participant NM



Note: BDI-II = Beck Depression Inventory–II; NSNE = Negative Self-Concepts and Negative Expectations.

variables that seemed to reliably influence depression outcome. Nonetheless, it might be worth noting that change was less robust across informants (i.e., the parents did not report as much change as the child) for the two males, who also presented with co-morbid disruptive behavior diagnoses. In addition, the two adolescents for whom activation appeared to be a mediator of treatment outcome were both from two-parent homes and had the shortest symptom durations on the DISC (4 and 6 months).

Discussion

An approach to conducting single-participant assessment of treatment mediators was outlined that requires documenting that the participant: (a) received treatment; (b) improved during treatment; (c) showed change on the mediator, which occurred at a reasonably expected time given the treatment protocol; and (d) preceded a substantial portion of the clinical improvement. Data from 4 depressed teens treated with VBBA were presented. Positive pretreatment change was identified for all 4 participants, 2 of whom went on to report additional symptomatic improvements very early in the course of treatment that were not attributable to VBBA. Onset of VBBA was associated with substantial changes in activation for 3 out of 4 participants, and for 2 this increased activation appeared to plausibly account for their within-treatment clinical gains. In no case did changes in dysfunctional thinking appear as a plausible mediator.

These data provide preliminary single-case support for the use of BA with depressed adolescents. Moreover, they provide some initial evidence that BA did, in the majority of cases, produce an increase in activation, and that this increased activation, in half of the cases, was followed by substantial changes in depressive symptoms. As such, these data generally coincide with the theoretically proposed mechanism of action in BA. Of course, the data from all of the participants did not correspond to predictions from the BA model. However, given the multitude of variables that may produce change during therapy (nonspecific factors, treatment specific factors, nontreatment-related factors), that 75% showed increased activation that corresponded to the introduction of VBBA, which for 50% preceded the changes in depression seems promising. Clearly, given the small sample size, the findings need to be replicated and are probably best treated as simply offering a hypothesis for testing in future research: 50% of responders to BA show a pattern of change consistent with the theorized mechanism.

The present findings provide not only some initial promising data for the use of BA with depressed youth but also suggest the potential promise of

the single-participant approach to assessing mediators. As such, we believe this approach warrants further research and development. Toward this end, several limitations and areas for future consideration are highlighted. To begin, it appears that certain data patterns may present some complications for the single-participant approach. For instance, cases where the effects of treatment are significantly delayed (i.e., sleeper effects) would make it hard to identify and make a clear inference about a mediator. For instance, if a client were taught cognitive restructuring but it didn't have an effect on negative cognitions until months of practice passed, or if the effects of changed cognitions on depressive symptoms did not occur for months, the temporal link between receipt of the treatment, change in cognition, and change in depression would be harder to detect. In addition, there would be a risk that a more proximal event would be seen as the agent that produced these changes. The current protocol provides some protection against misidentification because only one general therapeutic intervention was used; however, many protocols involve the use of multiple techniques/components, which would increase the risk for potential misidentification of the relevant treatment aspects producing change when that change was delayed.

A second potentially troublesome scenario for assessing mediators of treatment outcome at the single-participant level would be in the case where two mediators (e.g., ATQ and activity) change simultaneously and are followed by symptom change (e.g., a decrease in the BDI-II). Such a case would not allow for a clear inference to be made favoring one of these mediators over the other. Instead, both would have to be considered as plausible mediators, as would the possibility that a third variable produced the changes in both. The two patterns described, which could complicate the ability to draw clear inferences, might well need to be addressed in future research and could represent a significant limitation of the single-participant approach. However, before that can be said, it seems important to determine how often these complicating patterns occur. Future research using the single-participant approach would be useful in documenting the actual prevalence of these complicating scenarios. In addition, it should be further noted that it is not clear that single-participant assessment of treatment mediators is more vulnerable than group-level approaches to the complications raised by these data patterns; hence, the need for researchers to use multiple approaches to identifying mediators and to have results replicated both within and across methods.

When assessing mediators of treatment outcome at the level of the single-participant, we recommend adopting sample reporting conventions that mirror those used in clinical trials. That is, providing a description of the

number of potential participants, the number excluded from participation, the number who discontinued participation and when they discontinued, along with all relevant demographic and clinical characteristics. Provision of such information helps ensure that any data presented were not simply hand-picked to support a particular pattern or conclusion.

On the topic of sample size, it is useful to note that the approach described herein can be used with any sample size. In the present sample, 50% showed evidence for the activation mediator and 50% did not. Similar analyses could be done with a larger sample and the overall percentages reported. Graphs of the single-case data for all participants could not be presented in publications, due to space constraints, but the single-participant strategy could be employed and the data reviewed by a number of trained evaluators according to the criteria described. Inter-rater agreement could then be calculated to show that the evaluators' conclusions were similar, and then the group percentages reported (e.g., percentage of cases where the data supported a mediator). Given that the effects of therapy appear multiply determined, knowing the percentage of treatment responders whose results appear due, at least in part, to the targeted mechanism of action would be very useful. This information could then be used as a variable in other analyses to determine, for instance, if those whose change was plausibly due to the mechanism of action had lower relapse/recurrence rates. In addition, percentages could be compared across groups who received different treatments, such as those from a randomized clinical trial, to determine the percentage in each cell whose response was consistent with a proposed mechanism of action and whether this percentage differed across groups. The suggestion that individual-level analyses be applied to a large dataset has some precedence in, for example, Quitkin's (1999) approach to detecting placebo responders in antidepressant drug studies by conducting "pattern analyses" of individual participant data. The pattern analyses are based on session-by-session clinician ratings of either *improved* (1) or *not improved* (0) with certain patterns considered most likely to represent "true" responses to antidepressant drugs (e.g., 000111) and others placebo responses (e.g., 111111).

Identification of mediators at both the single case and group level require relatively brief measures of critical variables that can be administered repeatedly and maintain their reliability and validity (Kraemer et al., 2002). Recent data suggesting that weekly administrations of the BDI-II may produce greater positive changes than monthly or bimonthly use illustrates a potential problem (Longwell & Truax, 2005). Until a range of brief assessments that can be used repeatedly is fully available, use of the most widely studied measures seems prudent. Likewise, when possible, multiple

measures of a variable could be taken to look for convergence. For instance, the present study used a subscale of the widely studied ATQ as a repeated measure and, given the importance of activation, employed three measures of this variable. Furthermore, use of longer, psychometrically sound measures of the same variables at pretreatment and posttreatment can help determine whether the construct validity of the repeated measure was lost due to frequent administration. If the conclusion from a brief, repeated measure is inconsistent with the result of a more complete or alternative measure of the variable then confidence in the repeated measure's validity decreases. For instance, if a teen continued to meet DISC criteria for major depression at posttreatment but showed a consistent decline in the BDI-II across the course of treatment, there would be reason for concern that the BDI-II lost its construct validity. Incorporation of direct observation and/or performance-based measures (e.g., on laboratory tasks) should also occur whenever possible. In addition to these within-study considerations, replication of findings across studies is also critical.

In conclusion, single-participant assessment of treatment mediators appears to be a promising additional tool that can complement group-level analyses. Ideally, the two strategies should facilitate convergence on the mechanisms of action that are operative in successful psychotherapy. Existing criteria for evaluating the empirical support for a psychological intervention consider both between-group and single-participant data in drawing conclusions (Chambless et al., 1998). So too should evaluations of the mechanisms of action operating in those interventions. It is our hope that the present article provides a first approximation and encourages future research and methodological developments toward this end.

Note

1. Of note, a mediator may or may not ultimately be a mechanism. A mediator represents a correlational relationship, one plausible means via which a treatment exerts its effects, whereas a mechanism is considered causal. As such, mechanisms are inevitably mediators, but not vice versa (Kraemer, Wilson, Fairburn, & Agras, 2002).

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